Supplementary Information: Agentic Swarm Discovery of Single-Site Water Molecule

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1 Scope and authoritative model identity

This supplement describes the frozen single-site candidate c34 from the round-six campaign. It is distinct from the earlier three-site O/H/H model and earlier coarse-grained candidates. No diffusion, dielectric, phase-equilibrium, or caloric-closure result from those models is transferred to this candidate. All broad primary mean research gates pass, but the overall independent response assessment is **not qualified**: pressure depends materially on EOS representation and thermal-response sign and magnitude remain unresolved.

The authoritative parameter file is `model/frozen_c34.json`. Its SHA256 is

`b465820825ebda1337ac11ada2ac3942e1ed5ab287aed7c439a6d7cc3c29070c`.

Historical initialization metadata inside this byte-exact JSON does not describe the current fit. The companion freeze manifest, training records, and selection amendment provide current provenance. One bead represents one water molecule, with mass 18.01528 u; there are no explicit molecular bonds, angles, hydrogens, charges, or dipoles.

2 Complete symbolic definition and coefficients

The final model is

$$U = \sum_{i<j} \phi(r_{ij}) + \sum_i \Big[-0.1667931559207833 d_i - 2.8 d_i^2 + 0.2 d_i^4 + 5.999564839230267 \left[1 - 2n_i\right]_+^4 + 599.4640644 A_i \Big], \tag{1}$$

$$d_i = n_i - 2.536919437948505, \qquad n_i = \sum_{j \neq i} \left[1 - r_{ij}^2/5.5^2\right]_+^3, \tag{2}$$

$$A_i = \sum_{j<k} w(r_{ij}) w(r_{ik}) (\cos\theta_{jik} + 1/3)^2, \tag{3}$$

$$w(r) = t^3 Q(t)^2, \qquad t = \left[1 - r^2/4.3065^2\right]_+, \qquad Q(t) = \sum_{k=0}^{14} q_k t^k. \tag{4}$$

Energy coefficients are in kJ mol^{-1} and distances in Å. Coordination and weights are dimensionless. The potential is polynomial in invariant descriptors and piecewise polynomial in distances, rather than a global Cartesian polynomial.

k	q_k
0	-0.05584374858176977
1	13.813681810734714
2	-482.4017976897535
3	7165.51424376389
4	-57527.64697894118
5	297172.1111663983
6	-1072499.746376668
7	2814643.0354031
8	-5466093.281733246
9	7873236.650911203
10	-8316283.856326343
11	6264895.24457229
12	-3188503.053282442
13	982889.721893812
14	-138626.9692042507

The code uses Horner evaluation of Q and its derivative. Large alternating coefficients make cancellation relevant to excessively small finite-difference steps; the numerical audit is described below. The positive squared form guarantees nonnegative radial angular weights, not universal thermodynamic stability.

On interval $[a_j, a_{j+1})$ the pair energy is

$$\phi(r) = c_{3j}(r - a_j)^3 + c_{2j}(r - a_j)^2 + c_{1j}(r - a_j) + c_{0j}. \tag{5}$$

Below the first knot $a_0 = 1.8$ Å, the continuation is

$$\phi(r) = \phi(a_0) + \phi'(a_0)(r - a_0) + 10^4(r - a_0)^4. \tag{6}$$

Above the final knot the pair interaction is zero. The primary representation extends identically zero intervals to 6 Å solely for the neighbor representation; the physical pair support is approximately 4.3065 Å. The exact alias removes this redundant tail to 5.5 Å, preserving the broader coordination support. The printed table below is rounded; executable reproduction must use the full-precision JSON or exported parameters, not retype rounded coefficients. Coefficient dimensions are kJ mol^{-1} Å^{-p} for the coefficient multiplying the pth distance power.

Table 1: Pair cubic coefficients, printed to twelve significant digits. Full floating-point precision is retained in the accompanying JSON and CSV.

a_j	a_{j+1}	c_3	c_2	c_1	c_0
1.8	1.9	-479.245878163	509.310737482	-317.101890597	61.8215955623
1.9	2	-329.165370673	365.536974033	-229.617119445	34.7252679992
2	2.1	-218.796847395	266.787362831	-166.384685759	15.0897604244
2.1	2.2	-197.598829517	201.148308613	-119.591118615	0.900368629373
2.2	2.3	-35.8770813183	141.868659758	-85.2894217777	-9.24485897548
2.3	2.4	-80.1539441058	131.105535362	-57.9920022657	-16.390991637
2.4	2.5	-327.991094679	107.05935213	-34.1755135165	-20.959290454
2.5	2.6	126.344776082	8.6620237267	-22.6033759308	-23.6342393791
2.6	2.7	42.2350152808	46.5654565513	-17.080627903	-25.6816119588
2.7	2.8	41.7743141511	59.2359611355	-6.50048613429	-26.8817851683
2.8	2.9	-49.9034819405	71.7682553808	6.59993551734	-26.8976998562
2.9	3	-83.6238556898	56.7972107987	19.4564821353	-25.5699272326
3	3.1	-90.3507087974	31.7100540918	28.3072086243	-23.1399307668

a_j	a_{j+1}	c_3	c_2	c_1	c_0
3.1	3.2	-69.0549363765	4.60484145257	31.9386981788	-20.0824600722
3.2	3.3	67.8646824339	-16.1116394604	30.788018378	-16.9115967762
3.3	3.4	-22.0450786659	4.24776526978	29.6016309589	-13.9260466506
3.4	3.5	-22.1222266471	-2.36575832998	29.7898316529	-10.9454509807
3.5	3.6	-49.323297397	-9.0024263241	28.6530131875	-8.01224762531
3.6	3.7	-23.8137589737	-23.7994155432	25.3728290008	-5.2862938672
3.7	3.8	-21.5812735219	-30.9435432353	19.8985331229	-3.01081888153
3.8	3.9	22.4175304907	-37.4179252919	13.0623862702	-1.35198227511
3.9	4	48.1370775125	-30.6926661447	6.25132712656	-0.397505370516
4	4.1	52.6583729292	-16.2515428909	1.556906223	-0.0311622417938
4.1	4.2	4.33505274213	-0.454031012141	-0.113651167305	0.0146713245264
4.2	4.25	-5.41639037795	0.846484810497	-0.074405787469	0.00310095041666
4.25	4.28	5.78112398101	0.0340262538047	-0.0303802342539	0.000819824272215
4.28	4.3065	-7.9030395463	0.554327412095	-0.0127296242769	9.51312205101e-05
4.3065	5	0	0	0	0
5	5.5	0	0	0	0
5.5	6	0	0	0	0

3 Training, agent roles, and LLM-SR adaptation

Four collaborating contexts were used in the recent campaign: a coordinator, fitting/proposal specialist, native-engine specialist, and independent-validation specialist. Simulation processes and property evaluators are numerical tools, not additional reasoning agents. Historical specialist labels across the broader project must not be interpreted as simultaneous concurrency in this campaign. No ablation experiment establishes that collaboration itself improved search efficiency.

The actual upstream LLM-SR modules [1] were used at commit

`41c212312df6c16d936c9cb395356a62774c47e3`.

Agent-authored file continuations passed through native Sampler/Evaluator/ExperienceBuffer interfaces. The evaluator verified source-model hashes before registering cached measured fitness. Numerical coefficient fitting and MD were external. Seven intermediate feedback snapshots were inspected before adaptive proposals; the final native audit contains 40 records. The file replay verifies registered measurements and does not rerun MD. It is not an external API-based unattended search or exhaustive symbolic optimization.

There were 32 actual N64 five-state screens and eight N512 six-state refinements, covering 35 unique parameter sets. Screen states used 5 ps equilibration and 20 ps production; refinement states used 10 ps and 30 ps. All integration used 1 fs. The retained budget is 7.04 ns, comprising 5.00 ns N64 and 2.04 ns N512 across 250 trajectories, including diagnostics and failed/excluded work. The aggregate reported LAMMPS loop time of 2376.3 s accumulated over concurrent training is not OS CPU time or a controlled performance comparison.

Both an earlier coordination prototype and a polynomial mW-inspired seed were explored. No new AIMD force/energy dataset was used for this round. Trial families included independent radial-shell corrections, density-dependent positive angular strength, stabilized cubic/quartic embeddings, broader density descriptors, sparse low-coordination functions, and large radial-coordinate warps. The selected angular strength is constant and its embedding cubic coefficient is zero; these ablations are part of the result.

3.1 Selection amendment and held-out separation

The original strict same-budget minimax winner was c33, objective 0.935291, versus c34 at 0.972997. Both passed fitted mean research gates. An explicit pre-holdout amendment selected c34 because

both its global-plane and central-anchored training density predictions fell within the existing 0.5% density tolerance. The predictions were 0.993217 and 0.993647 g cm^{-3} for c34, versus approximately 1.004018 and 1.002712 for c33. This criterion was adopted after viewing training data, so the campaign is adaptive rather than a fully prespecified fixed-budget experiment.

The amendment, original winner, source hashes, and absence of fresh candidate trajectories at amendment time are recorded. Fresh testing uses primary c34 only; there is no retrospective swap to c33 if validation is unfavorable. Any new model revision must treat the present tests as training evidence and reserve another independent assessment.

3.2 Negative results and excluded measurements

A fast Born/fluctuation compressibility ranking disagreed with an available EOS check by approximately 68% and was excluded from fitness. Actual temperature–density EOS screens replaced it. Early static pressure/cohesion counterterms were proposal generators only: poor ensemble overlap and relaxation predictions were recorded. Pair-well compensation often stiffened the liquid. Large coordinate warps failed pressure or structure and were rejected.

A positional specification-lookup error reran parent models during an intended warped batch. Four affected EOS runs are preserved in a diagnostic directory, excluded from intended-family ranking, and replaced by correctly identified runs. All 40 selected-search records pass model identity/hash audits. These exclusions are evidence bookkeeping, not removal of unfavorable correctly identified candidates.

4 Fresh simulation protocols and references

Native LAMMPS [2] was tested locally with the 22 July 2025 runtime on macOS arm64. The engine is qualified here only for serial orthogonal periodic cells and real units. Candidate MD used 512 beads, 1 fs timesteps, and byte-identical frozen coefficients.

Ambient NPT: 298.15 K and 1 bar, 30 ps equilibration plus 150 ps production, thermostat damping 200 fs and barostat damping 2000 fs, sampling every 0.2 ps. Reserved seed: 227119. Five EOS NVT states: densities 0.98, 0.99705, and 1.015 g cm^{-3} at 298.15 K, plus 288.15/308.15 K at central density; 10 ps equilibration plus 60 ps production each, Langevin damping 5 ps^{-1}. Seeds: 226611, 226700, 226789, 226878, 226967. A separate RDF state used 295.15 K and 0.997772886443 g cm^{-3}, 10 ps plus 60 ps, seed 228119. Total fresh candidate integration: 600 ps.

Three NVT workers ran alongside one NPT process. This bounded dispatch changed neither seeds nor estimators; concurrent elapsed times are not benchmark results. Fresh trajectories start from declared geometries and independent velocities/noise; limited equilibration remains a scientific uncertainty.

The thermodynamic reference is IAPWS [3]. The ambient reference values used are density 0.997047039 g cm^{-3}, compressibility $4.524632587 \times 10^{-5}$ bar^{-1}, expansion $2.57287426 \times 10^{-4}$ K^{-1}, and thermal-pressure coefficient 5.6863717 bar K^{-1}. The cohesion target is 43.98748 kJ mol^{-1}, but the model diagnostic is a proxy. Heat-capacity references are 75.32763 and 74.53947 J mol^{-1} K^{-1}; their comparison with bead quantities does not restore missing molecular degrees of freedom.

The original Skinner numerical X-ray-derived oxygen RDF [4] at 22.0 Celsius was retrieved from the research mirror https://github.com/hoepfnergroup/LiquidStructureGP-Sullivan. The pinned commit and file are:

```
3a5df648a457a016bb07a954fef95c1b2d5feb1f
H2O/Experimental/Datasets/
Ambient_water_xray_data.txt
```

The data-file byte SHA256 is

`0af9f7d8e1ea73817b623177b65de030`
`5425a30f63cace33def3b138f5253960`.

The accompanying source manifest records URLs, hashes, and the mirror license. We used the original experimental columns, not the optional Gaussian-process reconstruction. The comparison integrates the piecewise-linear measured curve exactly with r^2 shell weighting in 0.05 Å bins. Estimated source errors were preserved without inventing bin independence or a full covariance matrix.

Native stock mW [5], with $\epsilon = 6.189$ kcal mol^{-1}, $\sigma = 2.3925$ Å, $a = 1.8$, $\lambda = 23.15$, and $\gamma = 1.2$, was run under matching static protocols. Its parameter conversion was audited. It is distinct from the polynomial mW approximation used as a search seed/control. Earlier OPC3 or round-five diffusion values are not c34 results.

5 Estimators and qualification

The locked pressure plane fitted actual state mean temperatures. It supplies the primary κ_T, α_P, and $(\partial P/\partial T)_\rho$ gates. NPT fluctuations and canonical covariance estimates remain separate consistency checks. In physical units,

$$\kappa_T = \frac{\langle (\Delta V)^2 \rangle}{k_B T \langle V \rangle}, \qquad \alpha_P = \frac{\langle \Delta V \Delta H \rangle}{k_B T^2 \langle V \rangle}. \tag{7}$$

The code consistently converts bar–Å^3 to kJ mol^{-1}. Configurational heat-capacity variances include the appropriate ideal translational contribution. They describe the coarse-grained Hamiltonian, not full molecular calorimetry.

NPT errors are descriptive ten-15-ps-block standard errors; fluctuation-response errors are delete-block jackknife estimates. EOS intervals use paired within-state pressure/temperature block resampling. The bootstrap does not certify coverage of slow correlations, equilibration error, or finite-size bias. Half-sampling values and nominal/actual-temperature sensitivity are retained.

Response	Central estimate	Descriptive 95% interval
κ_T (10^{-5}/bar)	4.435	[3.653,5.500]
α_P (10^{-4}/K)	1.366	[−3.333,5.817]
P_T (bar/K)	3.080	[−7.111,12.950]
Global-plane P_0 (bar)	66.45	[8.76,124.48]

The directly observed central pressure is 182.65 ± 59.63 bar; its temperature-anchored value is 180.17 bar and the quadratic value is 181.19 bar. Global, anchored, and quadratic roots are 0.994156, 0.989077, and 0.990265 g cm^{-3}. The five-parameter quadratic has no residual degrees of freedom; a density–temperature cross derivative cannot be identified. Its maximum primary plane residual/state-SE is 1.92, a descriptive ratio rather than a calibrated goodness-of-fit test.

The quadratic density curvature is approximately −508027 bar/(g cm^{-3})^2. Combining it with the measured NPT density variance gives a density-only Jensen correction near −55.12 bar. This is a plausible partial contributor to ensemble mismatch, not a full explanation or equilibrium proof. Applying the thermal quadratic to instantaneous kinetic-temperature fluctuations would be inappropriate because a thermodynamic state temperature is a different quantity.

NPT density is 0.995424 ± 0.000941 g cm^{-3}; its half means are 0.994628 and 0.996221. NPT pressure is 18.98 ± 7.62 bar. Independent NPT compressibility is $(4.111 \pm 0.409) \times 10^{-5}$/bar and expansion is $(1.022 \pm 0.730) \times 10^{-4}$/K. Effective $C_P = 36.019 \pm 1.994$ and $C_V = 35.035 \pm 1.293$ J mol^{-1} K^{-1}. The canonical thermal-pressure estimate, 7.809 ± 4.907 bar/K, is very noisy and does not replace the locked EOS estimate.

The gas reference is computed consistently:

$$E_{\text{iso}} = K_2 n_0^2 - K_1 n_0 + h_4 n_0^4 + A_4 = -3.3136661900\ mkJ\ mol^{-1}. \tag{8}$$

The cohesion proxy is 43.96791 ± 0.00559 kJ mol^{-1}. It is not a liquid–vapor coexistence calculation. Matched RDF full-range RMS is 0.123707 ± 0.001587 (descriptive delete-block SE), versus native mW 0.131820 ± 0.003003. First-shell RMS is 0.209377 ± 0.004202, versus mW 0.238453 ± 0.005671. Point-interpolation sensitivity for c34 full-range RMS is 0.123314. These sampling errors are distinct from experimental source uncertainty.

All nine broad primary mean gates pass, with positive EOS slope and worst normalized discrepancy 0.938128. Nevertheless the overall assessment is **not qualified**. The response intervals include zero, the pressure roots depend on representation, and half-trajectory pressure changes reach 259.51 bar. No automatic transport progression was performed. No source, model, or score was changed to exploit these held-out results.

6 Mechanical checks and exact optimizations

Candidate finite-difference force errors were below 10^{-5} kJ mol^{-1} Å^{-1}; native and analytic isolated-bead energies agreed. A 2 ps, 1 fs NVE test had per-bead drift about 1.16×10^{-5} kJ mol^{-1}. The fixed-strain virial difference at $h = 10^{-6}$ had absolute error 0.001194 kJ mol^{-1}, narrowly above the preset 0.001 limit. Coarser step convergence supported second-order truncation before small-step cancellation. Richardson estimates using $(10^{-4}, 3 \times 10^{-5})$ and $(3 \times 10^{-5}, 10^{-5})$ had errors 0.000563 and 0.000495, satisfying the unchanged tolerance. The original failure is preserved alongside the qualified numerical adjudication.

The first exact optimization removed zero pair intervals while retaining every nonzero radial, angular, and coordination edge. Its alias SHA256 is `8b8c18324246b64b0d9e147d2ec383f5d6240a9868c9ea0aef7030e184d4eb35`. Twenty training frames gave exactly zero energy, force, and six-component virial differences. The 2 ps alias NVE drift was 5.93×10^{-5} kJ mol^{-1} per bead.

The separate fused kernel accumulated coordination in the first neighbor pass and skipped zero radial work. An initial arithmetic reassociation exceeded its 10^{-8} parity tolerance by a small round-off amount and is retained as a failed diagnostic. Restoring the original angular arithmetic order yielded bit-exact twenty-frame parity. The 2 ps fused-kernel NVE drift was 2.27×10^{-6} kJ mol^{-1} per bead. The original primary source, binary, and Python runner remained unchanged. These checks qualify numerical equivalence for the tested frozen model, not arbitrary future parameters or production stability.

7 Controlled runtime protocols

Timing used Apple M2 arm64, one thread, 512 beads at density 0.99705 g cm^{-3}, 1 fs timesteps, common initial geometry, and matched initial velocity seeds per replicate. Each model used 3 ps Nose–Hoover NVT warmup at 298.15 K with 100 fs damping, followed by fresh setup and a single uninterrupted 10 ps NVE window. Identical native temperature/energy-drift sampling every 100 steps was included. There were no Python timestep callbacks or trajectory I/O within the timed

interval. Other campaign MD and substantial analysis jobs were stopped. All measured runs passed the short stability and mean-temperature-within-5% flags.

The four-model series used four Latin-rotation counterbalanced repeats. The separate fused-kernel series used three models and three rotations. These series are not pooled. Wall and LAMMPS loop timings, model/engine hashes, actual temperatures, energy drift, hardware, and thread settings are retained in JSON/CSV evidence.

Series	Implementation	Mean seconds/10 ps	Observed range
Four-repeat	Native mW	6.935	See original rows
	Polynomial mW	4.995	See original rows
	Primary c34	11.024	See original rows
	Cutoff alias	9.108	8.404–11.199
Three-repeat	Native mW	7.586	7.408–7.793
	Current alias	9.065	8.775–9.608
	Fused kernel	8.292	6.810–10.167

The alias reduced mean cost 17.38% relative to the primary in the first series, but was 31.33% more costly than native mW. The fused kernel reduced mean cost 8.53% relative to its matched alias control, but was 9.30% more costly than native mW. Ranges overlap and scheduling variability is large; there is no established native-mW speed advantage. The faster polynomial mW control does not establish the same benefit for the complete c34 model. No equal-accuracy, physical-transport-clock, larger-timestep, production MPI/GPU, or three-site-water speedup is inferred.

8 Reproduction boundary and remaining investigations

The compact Overleaf project includes TeX, figures, the exact primary JSON and radial CSV, selected evidence, and original native LAMMPS source/parameters/initialization. Compatible external LAMMPS headers, bindings, shared library, compiler, and plugin support are required. Runtime-specific binaries and large raw trajectories are omitted. A separate local review archive retains more campaign code and evidence; no public archival DOI is asserted. Full reanalysis of reported trajectory statistics requires the original raw trajectories in the campaign workspace.

The supplied native source is serial, orthogonal, real-units only. MPI decomposition, triclinic cells, restart serialization, per-atom stress, heat-current calculations, and production timestep scaling are unqualified. A separately rebuilt local library and the prepared 10 ps NVT input passed a reproduction smoke test; this is not additional property evidence. The reproduction notes retain their original campaign paths as provenance and require adapting paths after extraction. Parameters must include the many-body terms; a pair table alone is not this Hamiltonian.

Priorities are longer independent pressure/thermal replicas, intermediate-density EOS validation, and stationarity/size checks. A future model revision requires a new independent hold-out. Only then should diffusion and viscosity be calibrated jointly, followed by density-versus-temperature/TMD, interfaces, surface tension, vapor pressure, melting and boiling/coexistence, and sparse-environment clusters. A neutral scalar bead cannot intrinsically supply molecular dipole fluctuations or dielectric relaxation; a field-consistent response extension must be explicit. Missing molecular heat capacity requires separate treatment, not relabeling bead fluctuations as full-water calorimetry.

Language-model agents assisted the work and drafting. The named human author must review authorship, scientific claims, and any submission. The present project is an editable research manuscript, not a submitted or peer-reviewed article.

References

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